# CNTKNW

## Product Design Document

Research-backed product, UX, visual, and functional specification

### Version 0.2 - Canonical

Updated 24 July 2026

Prepared for human review and AI-agent implementation continuity.

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> **Source-of-truth rule:** Future prompts, design proposals, code changes, and AI-agent plans must surface any conflict with this document. Requirements may not be silently removed, weakened, renamed, or reinterpreted.

> **Scope of certainty:** This document is designed to remove ambiguity and implementation drift. It cannot guarantee that software will be built without defects. It defines the architecture, sequencing, tests, and guardrails required to make defects less likely and easier to detect.

# 1. Executive Summary

CNTKNW is a local-first personal knowledge museum and study studio. It combines serious source reading, visual analysis, journaling, drawing, vocabulary learning, and connected thinking without forcing the user to jump between a PDF reader, dictionary, whiteboard, notes app, and mind-map tool.

The product has only three primary destinations:

1. **Library** - books, papers, PDFs, collections, timelines, and the future immersive library.
2. **Gallery** - artworks, art history, comparison, visual annotation, and the future immersive museum.
3. **Notes** - complete journals, individual blocks, themes, glossary, connections, and visual maps.

Books and artworks open into one shared **Study Workspace**. The source appears beside one continuous journal. New text, quotations, drawings, questions, definitions, annotations, and links automatically inherit the active source context. While the user studies page 5, new journal material belongs to page 5 without asking the user to categorise it again.

CNTKNW's central advantage is not the number of features. It is the removal of context switching:

```
Open a source
-> read, look up, highlight, draw, question, connect, and reflect
-> close the source
-> return later to the exact intellectual context
```

# 2. Product Vision

## 2.1 Vision statement

CNTKNW is a place where a person's reading, visual study, vocabulary, questions, drawings, and connections remain attached to the sources that produced them while becoming part of one growing intellectual archive.

## 2.2 The problem

Existing tools split the study process:

- PDF readers preserve the source but treat thinking as comments in the margin.
- word processors support writing but lose source location and visual context;
- mind-map tools support spatial thinking but often duplicate or isolate the original notes;
- knowledge-management tools support linking but ask users to build and maintain elaborate systems;
- dictionaries and web searches break concentration and often lose the original passage;
- productivity timers count interaction rather than thought;
- beautiful note apps can become presentation tools rather than thinking tools.

CNTKNW should preserve flow while still creating durable, searchable structure.

## 2.3 Primary user outcomes

A user should be able to:

- import a large PDF and begin reading immediately;

- resume at the exact page, zoom, layout, and journal position;
- write or draw beside the source without deciding where the work belongs;
- inspect a book's complete journal in page order;
- connect page 5 to page 100, another book, an artwork, or a concept;
- define unfamiliar words offline without leaving the workspace;
- move between linear journal, outline, cards, and mind-map views without copying content;
- study an artwork with the same depth and continuity used for a book;
- review real study behaviour without being pressured to read faster;
- trust that their library still works when disconnected from the internet.

# 3. Product Principles

| Principle | Product rule |
|---|---|
| Source-aware by default | The active page, passage, artwork, region, game, or move supplies context automatically. |
| One canonical piece of content | Journal, outline, card, graph, and mind-map views render the same blocks rather than creating duplicates. |
| Think first, organise later | Capture must be immediate. Themes, links, types, and layout may be added during review. |
| Never leave the source to understand it | Dictionary, notes, annotations, search, and context remain inside the Study Workspace. |
| Local-first by design | Reading, writing, annotation, search, dictionary, and backup must work without an account or network. |
| Immersion plus speed | First-person spaces add discovery but never become the only route to content. |
| Depth over gamification | Statistics describe study; they do not reward rushing, punish rereading, or manufacture streak anxiety. |
| Progressive disclosure | A beginner sees a calm notebook. Advanced controls appear only when relevant. |
| Durable and portable | User data must be exportable, versioned, recoverable, and understandable outside the app where practical. |
| Visual restraint | Beauty comes from typography, rhythm, hierarchy, texture, and motion, not a rainbow of block colours. |
| Explicit relationships | A meaningful connection has an optional type and backlink, not merely a matching tag. |
| AI is optional and subordinate | Generated material must never replace source text or the user's interpretation silently. |

# 4. Research Synthesis: Why Users Adopt Thinking Tools

This section converts current productivity, journaling, PKM, whiteboard, and mind-map patterns into CNTKNW design decisions.

## 4.1 What successful products consistently do well

### Immediate capture

Users stay with tools that let them begin before they have designed an organisational system. Capacities explicitly uses daily notes as a low-friction inbox and encourages later review; Craft lets structure emerge by grouping blocks into pages. CNTKNW should provide the same freedom, but the active source location becomes the automatic inbox.

**CNTKNW decision:** opening a book and typing must create a valid page-linked block immediately. No mandatory title, category, tag, or template dialog.

### Atomic, movable units

Craft treats paragraphs and media as independent blocks. Tana turns nodes into typed objects through supertags. This flexibility allows content to gain structure later.

**CNTKNW decision:** all journal content is represented by stable, addressable blocks. A block can be moved, embedded, linked, transformed, or displayed in another view while retaining identity and history.

### One dataset, several useful views

Notion databases can display the same records as tables, lists, galleries, calendars, timelines, and charts. Xmind can switch between a visual map and an outline. These products succeed when a view changes presentation rather than creating a second copy of the data.

**CNTKNW decision:** Journal, Outline, Cards, Connections, and Mind Map are projections of the same blocks and relationships.

### Explicit objects and backlinks

Capacities reduces folder maintenance by treating books, people, topics, and other entities as object types. Backlinks reveal where an object was referenced and preserve context.

**CNTKNW decision:** Book, Artwork, Concept, Glossary Entry, Author, Artist, Period, and future Chess Game are typed entities. Every explicit link produces a backlink with source context.

### Spatial organisation without abandoning text

Heptabase organises note cards on whiteboards; Obsidian Canvas can place notes, media, PDFs, and text cards spatially; Miro combines mind maps, diagrams, notes, and other objects on a large canvas.

**CNTKNW decision:** spatial layout is a saved view over canonical blocks. It may add frames, positions, colours, and edges, but it must not own the only copy of important writing.

### Ownership and offline trust

Obsidian's local files and Anytype's local-first architecture appeal to users who care about offline access, privacy, and long-term ownership.

**CNTKNW decision:** core capability has no CDN or cloud requirement. Binary assets, notes, dictionary packs, indexes, and backups are local. Sync can be added later as an optional layer.

### Visual polish that requires little design skill

Craft emphasises beautiful defaults, typography, covers, cards, and media without requiring users to become designers. Users adopt polished tools because their work feels valuable immediately.

**CNTKNW decision:** provide excellent default typography and block treatments. Customisation should be bounded by a coherent design system rather than unlimited per-block decoration.

### Focus modes

Xmind provides Zen mode and hides peripheral controls. Serious writing and reading tools create calm by removing chrome instead of adding motivational clutter.

**CNTKNW decision:** Source Focus, Journal Focus, Deep Reading, and Presentation modes should be first-class workspace states.

## 4.2 Repeated sources of frustration

Community discussion and product behaviour reveal several predictable failure modes:

- quick capture becomes slow when users must choose folders, vaults, schemas, or properties first;
- extensive plugins and configuration create a powerful tool that feels like a project to maintain;
- beautiful databases become clumsy for tiny thoughts or mobile capture;
- graphs become decorative noise when links are created for volume rather than meaning;
- canvas-only notes become hard to search, reuse, export, or review linearly;
- cloud dependence weakens trust and offline reliability;
- separate mobile and desktop interaction models can make a system feel inconsistent;
- sync conflicts and invisible state changes damage confidence;
- too many block types or colours make a journal look fragmented rather than expressive;
- AI-generated structure can overwhelm personal thinking and make provenance unclear.

## 4.3 What CNTKNW should adopt

| Pattern | Adopted form in CNTKNW |
|---|---|
| Craft-style atomic blocks | Stable JournalBlock records with rich presentation and nesting. |
| Capacities-style typed objects | Sources, people, concepts, glossary terms, periods, and future games. |
| Tana-style later typing | A plain text block may later become a question, claim, definition, or concept without replacement. |
| Notion-style multiple views | The same source collection or blocks shown as list, timeline, cards, journal, or map. |

| Pattern | Adopted form in CNTKNW |
|---|---|
| Heptabase-style spatial synthesis | Saved canvases made from canonical blocks and sources. |
| Obsidian-style backlinks and ownership | Contextual backlinks, local data, export, and no core cloud dependency. |
| Xmind outline-map duality | One-click switching between hierarchical outline and mind map. |
| Miro-style direct manipulation | Drag, connect, group, frame, zoom, and navigate naturally on canvases. |
| Craft-style visual defaults | Editorial typography, strong spacing, premium motion, and visually legible block types. |

### 4.4 What CNTKNW should avoid

- A blank graph as a headline feature before meaningful links exist.
- A separate whiteboard database that duplicates journal content.
- A slash menu with dozens of equal-priority choices.
- Mandatory folder or tag decisions during reading.
- A plugin-dependent core experience.
- A first-person world that makes opening a book slower.
- Flattened PNG-only drawings that cannot be edited.
- Cloud-hosted PDF workers or dictionary lookups in an offline product.
- Colour-coding every block type so aggressively that a journal resembles a dashboard.
- AI that writes into journals without a visible origin, review state, or user action.

## 5. Information Architecture

### 5.1 Three primary destinations

```
CNTKNW
|
+-- Library
|   +-- Collection view
|   +-- Timeline view
|   +-- Search and quick open
|   +-- Immersive library (later)
|   +-- Book/PDF -> Study Workspace
|
+-- Gallery
|   +-- Collection and timeline views
|   +-- Search, rooms, artists, periods
|   +-- Immersive museum (later)
|   +-- Artwork -> Study Workspace
|
+-- Notes
    +-- Journals
    +-- All blocks
    +-- Themes and concepts
    +-- Connections and backlinks
    +-- Glossary
    +-- Maps and canvases
    +-- Standalone notes
```

### 5.2 Secondary global tools

The following live in contextual menus, overlays, or settings rather than permanent main tabs:

- universal search and command palette;
- timer and session controls;
- recent and resume;
- dictionary pack management;
- import, export, and backup;
- application settings;
- future Chess Room access through the Library world.

### 5.3 Primary navigation requirement

**IA-001:** No new primary tab may be added without an explicit product-design revision. A coding agent may not promote Sessions, Whiteboard, PDF Reader, Dictionary, Export, Settings, or Chess to permanent top-level destinations.

# 6. Shared Study Workspace

The Study Workspace is the central interaction surface for books and artworks.

```
+--------------------------------+--------------------------------+
| SOURCE PANE                    | JOURNAL PANE                   |
|                                |                                |
| PDF or artwork                 | Current context header         |
| Text/visual annotation layer   | Page-linked journal blocks     |
| Search, contents, navigation   | Drawings, maps, concepts       |
| Zoom, view mode, compare       | Backlinks and connections      |
|                                |                                |
+--------------------------------+--------------------------------+
| Source location | Focus 43:21 | Autosaved | Dictionary | View |
+-----------------------------------------------------------------+
```

### 6.1 Workspace states

| State | Behaviour |
|---|---|
| Split | Source and journal visible side by side. |
| Source focus | Source uses most of the screen; journal remains one action away. |
| Journal focus | Journal expands for long-form writing while location remains visible. |
| Compare | Two sources or two source locations appear with one journal context. |
| Deep reading | Non-essential chrome hides; timer and location remain subtle. |
| Review | Complete source journal opens in reading order with source previews. |

### 6.2 Resume contract

For every source, CNTKNW must persist:

- exact source location;
- zoom and view mode;
- source and journal scroll positions;
- split-panel dimensions and collapsed state;
- active journal block and text selection where feasible;
- selected annotation tool;
- open dictionary or connection panel;
- last active timestamp.

**WS-001:** reopening a source restores its previous workspace state rather than resetting to page 1.

### 6.3 Save behaviour

- Every durable edit autosaves.
- A visible state communicates `Saving`, `Saved locally`, `Backup needed`, or `Write failed`.
- Related changes use transactions where possible.
- Failure never clears the editor.
- Recovery snapshots protect against crashes and accidental deletion.

# 7. The Infinite Source-Linked Journal

### 7.1 Locked behaviour

Each source has one continuous journal. Journal content is grouped by populated source locations.

While PDF page 5 is active:

- a paragraph belongs to page 5;

- a quote belongs to page 5 and its selected text range;
- a drawing belongs to page 5;
- a question belongs to page 5;
- a definition belongs to page 5;
- a connection created from the passage begins at page 5.

Changing to page 6 changes the active context. It does not erase page 5 and does not create an empty page 6 section until content is added.

## 7.2 Canonical block model

A JournalBlock is the atomic unit. Every block has:

- stable ID;
- journal ID;
- block type;
- source location;
- canonical content payload;
- order and optional parent block;
- created and updated timestamps;
- provenance and author type;
- version/revision metadata;
- optional concept and connection references.

## 7.3 Block types

| Type | Purpose | Default appearance |
|---|---|---|
| Paragraph | Long or short reflection. | Quiet body text. |
| Heading | User-created structure. | Editorial serif or strong sans heading. |
| Quote | Captured source text. | Indented paper card with source marker. |
| Question | Unresolved inquiry. | Subtle question marker; not a loud coloured box. |
| Claim | Proposition the user is advancing. | Strong left rule and label. |
| Evidence | Material supporting a claim. | Compact source-aware block. |
| Objection | Challenge to a claim. | Contrasting rule and relationship field. |
| Definition | General, source-specific, or personal meaning. | Term-forward dictionary card. |
| Drawing | Editable vector sketch or diagram. | Embedded canvas with preview. |
| Argument map | Structured claims and typed logical edges. | Embedded graph with outline fallback. |
| Image | Imported or captured visual. | Caption and source controls. |
| Connection embed | Preview of another block, page, source, or concept. | Compact backlink card. |
| Divider | Visual rhythm. | Thin editorial separator. |
| Callout | Deliberately emphasised note. | Restrained neutral surface. |

## 7.4 Capture menu

The primary + menu is ordered by likely study action:

1. Write
2. Quote selection
3. Question
4. Draw
5. Define
6. Connect
7. More: claim, evidence, objection, image, map, timeline, comparison

Typing immediately creates a paragraph. The user never has to choose a type first.

## 7.5 Transform rather than recreate

A paragraph can later be transformed into a question, claim, definition, card, or concept while retaining:

- the same ID;
- backlinks;
- source location;
- revision history;

- comments or relationships;
- placement on maps.

# 8. Journal Views and Mind Maps

## 8.1 Same information, different views

| View | Best for |
|---|---|
| Journal | Reading the user's thought process in source order. |
| Outline | Understanding hierarchy and collapsing detail. |
| Cards | Reviewing selected insights, quotes, concepts, or questions. |
| Connections | Inspecting explicit links and backlinks with context. |
| Mind Map | Developing a central concept through branches. |
| Free Canvas | Spatial synthesis across unrelated or cross-source material. |
| Argument Map | Testing support, objection, contradiction, and conclusion. |

## 8.2 Mind-map architecture

A mind map must not duplicate note text. It stores a layout record that references canonical nodes.

```
MindMapView
  id
  title
  rootNodeRef
  nodePlacements[] -> JournalBlock | Concept | SourceLocation | Source
  edgePresentations[] -> Connection or view-only hierarchy
  frames[]
  viewport
  collapsedNodeIds[]
  stylePreset
```

## 8.3 Automatic map generation

CNTKNW can generate an initial map from:

- a concept and its direct connections;
- a page block and related pages;
- a book's themes;
- an argument block hierarchy;
- selected journal blocks;
- a search result set.

The generated layout is a starting point, not a permanent automatic rearrangement. Once the user edits positions, CNTKNW preserves the personal spatial layout.

## 8.4 Map interaction requirements

- drag nodes without altering canonical content order;
- edit a node in place and update the original block;
- expand and collapse branches;
- open source context beside the map;
- create typed connections by dragging ports;
- select multiple nodes and group them in a frame;
- zoom to selection, fit view, use minimap, and navigate by keyboard;
- switch the same map to outline view;
- distinguish canonical connections from view-only parent-child layout edges;
- avoid constant physics motion after placement;
- export an image while preserving an editable native format.

## 8.5 Argument maps

Argument maps use explicit semantics:

- supports;

- objects to;
- rebuts;
- qualifies;
- depends on;
- concludes;
- example of;
- definition of.

The map must also be reviewable as a nested textual outline for accessibility and long-form reasoning.

# 9. Visual and Interaction Design Bible

## 9.1 Desired feeling

CNTKNW should feel like a quiet private museum, an archival reading room, and a researcher's notebook - not a corporate project dashboard.

Keywords:

- scholarly;
- intimate;
- tactile;
- collected;
- calm;
- precise;
- premium;
- lived-in without becoming messy.

## 9.2 Visual hierarchy

Beauty should come from:

- generous but efficient spacing;
- strong typographic contrast;
- consistent alignment;
- legible text measure;
- subtle surfaces and borders;
- carefully cropped covers and artwork;
- meaningful density changes;
- restrained animation;
- context-aware previews.

## 9.3 Typography

Recommended system:

- **UI sans:** Inter, Geist, or an equivalent highly legible variable sans.
- **Editorial serif:** Source Serif 4, Literata, or an equivalent for journal headings, quotations, and reading-oriented display.
- **Monospace:** a legible code/data face for citations, identifiers, and technical excerpts.

Rules:

- body journal measure targets roughly 60-75 characters per line;
- source text and journal text have independent zoom controls;
- minimum default body size should remain comfortably readable on a laptop;
- headings use scale and spacing before decorative colour;
- italics and low-contrast text are not used for essential information.

## 9.4 Colour

Use a neutral dark foundation with warm paper-like reading surfaces and one primary accent. Context may have subtle secondary accents, but blocks should not become a rainbow.

Suggested semantic roles:

| Token role | Purpose |
|---|---|
| Canvas | Deep neutral application background. |

| Token role | Purpose |
|---|---|
| Surface | Panels and toolbars. |
| Paper | Reading and journal writing plane. |
| Ink | Primary text. |
| Muted ink | Metadata and secondary labels. |
| Accent | Selection, focus, links, active tools. |
| Success | Saved and restored states. |
| Warning | Backup, storage, or inactivity attention. |
| Danger | Destructive action only. |

### 9.5 Block aesthetics

- Paragraphs remain mostly unboxed.
- Quotes use indentation, source markers, and subtle paper contrast.
- Questions use a small icon or label rather than a saturated card.
- Definitions prioritise the term and layer of meaning.
- Connections show direction and relationship label.
- Drawings and maps have minimal chrome until hovered or focused.
- Page/location headers are sticky, compact, and collapsible.
- Repeated cards use consistent corner radius and shadow depth.

### 9.6 Motion

Motion should explain state:

- a new block unfolds and focuses the cursor;
- a source selection becomes a quote card through a short spatial transition;
- opening a backlink reveals its origin rather than teleporting without context;
- panel resizing and collapse use quick, non-bouncy easing;
- autosave indicators fade rather than flash;
- reduced-motion preference disables non-essential transitions.

### 9.7 Iconography

Use one consistent outlined icon family. Icons should support labels until the interaction is familiar. Do not mix emoji, text symbols, and unrelated icon styles in production UI.

### 9.8 Empty states

Empty states should teach one next action:

- Library: `Import your first PDF` with a sample-book option.
- Page journal: `Start writing, quote the source, or draw.`
- Connections: explain that links appear only when meaningful.
- Mind map: invite selection of existing blocks rather than showing a meaningless demo graph.

## 10. Onboarding and Adoption

### 10.1 First-session path

1. Choose local library location or browser storage mode.
2. Import a PDF or open a built-in sample.
3. CNTKNW opens the Study Workspace directly.
4. A three-step contextual guide demonstrates write, quote, and return-to-source.
5. The app saves automatically and shows where data lives.
6. Mind maps, concepts, advanced annotations, and 3D are introduced later through use, not an upfront tour.

### 10.2 Adoption principles

- deliver value before asking for customisation;
- provide excellent defaults;

- reveal storage and backup status clearly;
- offer templates as optional starting structures, not mandatory systems;
- preserve common keyboard conventions;
- make undo reliable and visible;
- show a recent/resume surface on launch;
- use examples connected to the user's source, not generic productivity examples.

### 10.3 Beginner and advanced modes

Do not create two separate products. Use progressive disclosure:

- beginner sees writing, quote, question, draw, define, connect;
- advanced user can enable claims, argument semantics, custom block views, map controls, metadata editing, and automation.

## 11. Library

### 11.1 Core 2D Library

Required before 3D:

- search and quick open;
- recent and resume;
- shelf/card/list views;
- categories and collections;
- authors, dates, subjects, and reading state;
- timeline with selectable date dimension;
- import, archive, and metadata correction;
- per-book statistics and journal preview.

### 11.2 PDF import

Metadata precedence:

1. embedded PDF metadata;
2. title-page text extraction;
3. cleaned filename;
4. user correction.

Extract or infer where available:

- title;
- author;
- page count;
- subject and keywords;
- table of contents;
- creation/publication dates;
- language;
- cover thumbnail.

### 11.3 Reading modes

- continuous vertical scroll;
- single page;
- later two-page spread;
- deep reading/focus mode;
- side-by-side comparison;
- accessible text extraction view when available.

### 11.4 Future immersive library

- first-person WASD navigation;
- mouse and keyboard look controls;

- readable spines and covers;
- rooms generated from categories, eras, authors, or collections;
- direct interaction opens the same Study Workspace;
- search/teleport and 2D fallback always available;
- performance settings and reduced-motion options;
- Chess Room placed as an optional side room.

# 12. Gallery

## 12.1 Core Gallery

- artwork import and high-resolution storage;
- artist, title, date, period, movement, culture, location, medium, dimensions, and provenance metadata;
- timeline, location, collection, and comparison views;
- search and filters;
- favourite, studied, and collection states.

## 12.2 Artwork Study Workspace

- zoom, pan, fit, and high-resolution detail;
- non-destructive region and drawing annotations;
- region-linked journal blocks;
- optional analysis stages: observation, formal analysis, interpretation, context, evaluation, personal response;
- side-by-side artwork comparison;
- links to passages, concepts, artists, periods, and other works.

## 12.3 Future immersive museum

- walkable rooms organised by period, location, movement, or collection;
- artwork scale and spacing chosen for viewing rather than decorative crowding;
- room labels and contextual plaques;
- immediate switch to 2D collection or Study Workspace;
- accessibility route that does not depend on first-person navigation.

# 13. Notes

## 13.1 Notes views

| View | Contents |
|---|---|
| Journals | Complete book or artwork journals in source order. |
| All blocks | Filterable atomic content across sources. |
| Themes | Concepts, tags, and related material. |
| Connections | Explicit links, backlinks, and relationship filters. |
| Glossary | Looked-up words, source meanings, personal definitions. |
| Maps | Saved mind maps, argument maps, and canvases. |
| Standalone | Notes created before or outside a source. |

## 13.2 Filters

- source type;
- specific book or artwork;
- page, passage, or region;
- block type;
- theme or concept;
- relationship type;
- author or artist;
- date created, edited, or studied;
- saved vocabulary;

- questions unresolved or resolved.

### 13.3 Review behaviour

Review should support:

- expanding source preview beside a block;
- jumping to exact passage or visual region;
- turning an unstructured block into a concept;
- merging duplicate concepts safely;
- connecting distant pages;
- selecting blocks and creating a map;
- preserving the original journal order after thematic grouping.

## 14. Connections and Knowledge Model

### 14.1 Connection endpoints

A connection can target:

- source;
- source location;
- journal block;
- annotation;
- concept;
- glossary entry;
- author or artist;
- artwork region;
- future chess game, move, or position.

### 14.2 Relationship types

Initial set:

- same theme;
- supports;
- contradicts;
- develops;
- resolves;
- example of;
- influenced by;
- question answered by;
- visually represents;
- comparison with;
- custom.

### 14.3 Link quality

CNTKNW should encourage `link for meaning, not for volume`:

- unlinked mentions may be suggested separately;
- explicit backlinks represent deliberate relationships;
- graph defaults filter weak metadata links;
- repeated automatic links do not flood the journal.

## 15. Offline Dictionary, Glossary, and Concept Cards

### 15.1 Dictionary interaction

Selecting or double-clicking a word may show:

- definition;

- part of speech;
- pronunciation;
- alternative senses;
- synonyms and antonyms;
- etymology;
- usage examples;
- related words.

The user may:

- close without saving;
- save the lookup;
- add a personal meaning;
- define how the source uses the term;
- create or link a concept;
- view occurrences in the current source or library.

### 15.2 Three layers of meaning

1. **Dictionary meaning** - general language definition.
2. **Source meaning** - how a book, author, period, or discipline uses the term.
3. **Personal concept** - the user's evolving interpretation and cross-source connections.

### 15.3 Offline packs

- primary English dictionary pack;
- optional philosophy terminology;
- art-history terminology;
- literary terminology;
- theology and mythology;
- historical-period packs;
- future language packs.

Licences, attribution, update mechanism, package size, and search index format must be documented before distribution.

## 16. PDF and Annotation Requirements

### 16.1 PDF requirements

- core rendering works offline;
- worker assets are bundled locally;
- large PDFs render incrementally;
- text layer supports selection and search;
- page thumbnails load lazily;
- page and workspace state persist;
- table of contents is retained or derived where possible;
- original binary remains unmodified;
- scanned PDFs may later use optional local OCR.

### 16.2 Annotation model

Annotations are separate from PDF binary content and use normalised coordinates.

Supported types:

- text highlight;
- underline;
- strikethrough;
- freehand pen;
- rectangle and ellipse;
- arrow and line;
- text callout;
- region selection;

- bookmark.

Every annotation stores:

- source and asset IDs;
- page;
- type;
- normalised geometry;
- style;
- optional exact text anchor and quote;
- linked journal block IDs;
- timestamps and revision.

### 16.3 Drawing requirement

Editable drawing data is canonical. A raster preview may be cached, but CNTKNW must not store only a flattened PNG.

## 17. Search and Retrieval

### 17.1 Universal search

Search across:

- book and artwork metadata;
- journal text;
- block titles and types;
- glossary and concepts;
- explicit connections;
- extracted PDF text;
- captured quotes;
- annotation callouts;
- future OCR text.

### 17.2 Search experience

- instant command palette for navigation and actions;
- result grouping by source and type;
- passage snippets with highlighted terms;
- keyboard navigation;
- filters without leaving the result page;
- exact return to source location;
- saved searches as optional review views;
- no network requirement for core index.

## 18. Timer, Sessions, and Statistics

### 18.1 Timing modes

| Mode | Behaviour |
|---|---|
| Manual focus | Default. Counts until the user pauses or ends. Quiet rereading counts. |
| Assisted auto-pause | Optional. Detects inactivity and asks before removing time. |
| 25/5 | Classic Pomodoro. |
| 50/10 | Deep reading default. |
| Custom | User-defined focus, breaks, and cycles. |
| Stopwatch | Open-ended manual session. |

Activity includes reading interactions, page navigation, dictionary use, search, writing, drawing, annotation, and connection work. Lack of interaction is not proof of inactivity.

### 18.2 StudySession

A session captures:

- source type and ID;
- start and end times;
- focused, break, manual-pause, and removed-idle seconds;
- start and end source locations;
- locations visited;
- blocks, annotations, dictionary lookups, and connections created;
- timer mode;
- optional reflection.

### 18.3 Statistics

Book statistics:

- focused time and sessions;
- current and highest page;
- pages studied and revisited;
- average and longest session;
- journal blocks, quotes, questions, drawings, and connections;
- vocabulary and concepts;
- page-range activity heatmap.

Artwork statistics:

- viewing and analysis time;
- revisits;
- observations, annotations, questions, and connections;
- analysis stages completed;
- comparison activity;
- artists, periods, movements, and media explored.

Stats are descriptive and may be hidden. They do not score the quality of thought.

## 19. Local-First Storage, Privacy, and Backup

### 19.1 Storage principles

- domain records and binary assets are separated;
- schemas are versioned;
- every schema change has a migration;
- core actions work offline;
- component code does not write directly to arbitrary object stores;
- local full-text indexes can be rebuilt from canonical data;
- storage quota and write errors are visible;
- future sync is optional and cannot become the only source of truth.

### 19.2 Backup levels

| Backup | Contents |
|---|---|
| Notes-only | Metadata, journals, concepts, glossary, connections, sessions, settings. |
| Portable library | Notes plus PDFs, artwork, annotations, editable drawings, indexes manifest. |
| Incremental archive | Changes since the last snapshot. |
| Markdown export | Human-readable journals and metadata with stable link references. |

### 19.3 Recovery

- block revision history;
- soft deletion and trash;
- database transaction recovery;
- import validation before writes;

- conflict report rather than silent overwrite;
- backup verification and restore test;
- visible last-successful-backup status.

# 20. Accessibility and Device Behaviour

## 20.1 Accessibility

- all core actions keyboard-accessible;
- semantic headings and landmarks;
- screen-reader text alternatives for maps and drawings;
- visible focus states;
- sufficient contrast;
- colour never carries meaning alone;
- reduced-motion support;
- resizable text and panels;
- alt text and captions for artwork;
- outline alternative for every mind or argument map.

## 20.2 Responsive strategy

Desktop/laptop is the complete experience. Tablet supports reading, annotation, drawing, and review. Phone focuses on capture, reading, dictionary, recent items, and lightweight review rather than forcing a miniature desktop canvas.

# 21. Future Chess Room

## 21.1 Scope

Chess is a later optional room inside the immersive Library. It is not a fourth primary tab and is not required for the core product release.

Modes:

- break;
- casual play;
- lessons;
- scenarios and puzzles;
- game review;
- historical-game archive.

## 21.2 Bot personas

Bots should combine:

- target rating band;
- search limits;
- error distribution;
- opening repertoire;
- time behaviour;
- style: tactical, positional, aggressive, defensive, endgame-focused, unpredictable;
- optional teaching behaviour.

Initial production versions should use original fictional Library characters. Real living players, photographs, voices, recognisable likenesses, names, or implied endorsements require appropriate permission and asset rights. Licensed grandmaster packs may be added later.

## 21.3 Knowledge integration

Chess notes may attach to games, moves, positions, openings, lessons, and personal mistakes. A chess book may open with the PDF on one side and an interactive board beside the journal. Chess play, chess study, and reading-break time remain distinct in statistics.

# 22. Data Model Overview

## 22.1 Core entities

```
Source
  BookSource
  ArtworkSource
  Future ChessSource

Asset
  PdfAsset
  ImageAsset
  DictionaryPack

WorkspaceState
Journal
JournalBlock
Annotation
Connection
Concept
GlossaryEntry
MindMapView
StudySession
Collection
Person / Creator
```

## 22.2 Universal source location

```
type SourceLocation = {
  sourceType: 'book' | 'artwork' | 'chess-game' | 'chess-position';
  sourceId: string;
  pdf?: { page: number; textAnchor?: TextAnchor };
  artwork?: { region?: NormalizedRect };
  chess?: { moveNumber?: number; fen?: string };
};
```

## 22.3 Page sections are computed

A page section is not a second copy of journal content. It is computed by grouping blocks with matching source IDs and PDF page numbers. Only groups containing visible blocks appear.

# 23. Non-Functional Requirements

| ID | Requirement |
|---|---|
| NFR-001 | Core reading, editing, annotation, dictionary, and search function offline. |
| NFR-002 | Autosaved text should survive ordinary application crashes and refreshes. |
| NFR-003 | Opening a large PDF must not load every rendered page into memory. |
| NFR-004 | Scrolling and typing should remain responsive during indexing and autosave. |
| NFR-005 | All schema changes include deterministic migrations and rollback/recovery instructions. |
| NFR-006 | No core runtime dependency is fetched from a CDN. |
| NFR-007 | Search index can be rebuilt from canonical records. |
| NFR-008 | Every destructive action has undo, trash, confirmation, or an equivalent recovery path. |
| NFR-009 | The application reports storage, import, export, and backup failures clearly. |
| NFR-010 | Core user data can be exported without a subscription or network connection. |

# 24. Delivery Roadmap

| Phase | Scope | Exit condition |
|---|---|---|
| 0. Foundation reset | TypeScript, routing, design tokens, repository layer, schemas, migrations, tests. | App shell builds; no direct component DB writes; legacy data can be read. |
| 1. Reliable PDF core | Import, metadata, local worker, text layer, search, resume, lazy rendering. | Large PDF opens offline and resumes exactly. |
| 2. Infinite journal | Canonical blocks, page context, autosave, block editor, journal review. | Page 5 work always returns under page 5. |
| 3. Annotation and drawing | Text highlights, vector overlays, editable drawing blocks, exact source links. | Annotations survive zoom, mode changes, and reopen. |
| 4. Connections and concepts | Typed links, backlinks, concept cards, relationship review. | Distant pages and sources connect bidirectionally. |
| 5. Dictionary and glossary | Offline pack, lookup, source meaning, personal meaning, occurrences. | A difficult word can be understood and saved offline. |
| 6. Notes, search, and maps | Global views, local index, outline, cards, mind map, argument map. | Same blocks switch views without duplication. |
| 7. Artwork study | Gallery metadata, viewer, analysis, annotations, comparisons. | Artwork uses the shared workspace model. |
| 8. Sessions and statistics | Manual timer, assisted pause, sessions, source stats. | Timing is transparent and source-aware. |
| 9. Production polish | Accessibility, performance, recovery, onboarding, backups, responsive layouts. | Daily-use release candidate. |
| 10. Immersive worlds | Library/museum 3D scenes and 2D fallback. | 3D reads stable catalogue APIs only. |
| 11. Chess Room | Bots, break integration, lessons, scenarios, review. | Optional expansion after core stability. |

# 25. Acceptance Criteria

| ID | Acceptance criterion |
|---|---|
| AC-001 | Importing a PDF creates or attaches a book and opens it without requiring the network. |
| AC-002 | Reopening a book restores its exact page, zoom, panel state, and journal context. |
| AC-003 | Text, drawing, quote, question, or definition created while page 5 is active inherits page 5 automatically. |
| AC-004 | Navigating to page 6 creates no visible empty journal section. |
| AC-005 | Page 5 may link to page 100 with a typed relationship and both endpoints show context-rich backlinks. |
| AC-006 | A selected word can be defined offline, saved, given a personal meaning, and reopened at the original passage. |
| AC-007 | PDF annotations remain aligned after zoom, resize, reopen, and view-mode change. |
| AC-008 | Editable drawings retain vector/object data rather than only raster output. |
| AC-009 | A block edited from a mind map updates the same block in the journal. |
| AC-010 | Switching between journal, outline, cards, and map creates no duplicate content records. |
| AC-011 | Manual timing continues through quiet rereading until the user pauses or ends it. |
| AC-012 | Assisted auto-pause asks before removing time unless immediate pause is explicitly enabled. |
| AC-013 | Notes-only and full portable-library backups validate and restore successfully. |
| AC-014 | Library and Gallery remain fully usable without 3D navigation. |
| AC-015 | Core actions are keyboard-accessible and maps have outline alternatives. |
| AC-016 | Chess may be disabled and never contaminates reading or art statistics. |

# 26. Explicit Non-Goals for the Core Release

- team collaboration and multiplayer editing;
- public social network or content feed;
- a universal project-management replacement;

- cloud-only sync;
- mandatory AI summaries;
- automatic evaluation of philosophical correctness;
- immersive 3D before the 2D core is reliable;
- real-player chess likenesses without licensing;
- a plugin marketplace before stable extension contracts exist;
- full desktop/mobile feature parity in the first release.

# 27. Open Product Decisions

- final brand stylisation and pronunciation;
- web-first installable PWA versus desktop-first Tauri packaging;
- browser IndexedDB library versus user-selected filesystem library for the first production release;
- Tiptap/ProseMirror versus another extensible block editor after a short technical spike;
- exact vector-canvas library shared between PDF overlays and journal drawings;
- OCR inclusion in the first public release;
- dictionary dataset and licence;
- default immersive art direction;
- optional local or cloud AI scope;
- public product, private personal tool, or both;
- whether licensed chess partners become a commercial expansion.

# 28. Research References

The recommendations above were derived from official product documentation and current community discussion. Product features can change; implementation agents should verify current APIs before choosing dependencies.

**1.** Obsidian Help - Canvas and JSON Canvas: https://obsidian.md/help/plugins/canvas

**2.** Capacities Documentation - Getting started, objects, backlinks, networked notes, and daily notes: https://docs.capacities.io/tutorials/getting-started

**3.** Capacities Documentation - Backlinks: https://docs.capacities.io/reference/backlinks

**4.** Craft Help Center - Blocks and Pages: https://support.craft.do/en/write-and-edit/blocks-and-pages

**5.** Notion Help Center - Intro to databases and multiple views: https://www.notion.com/help/intro-to-databases

**6.** Anytype Documentation - local-first and user-controlled data: https://doc.anytype.io/anytype-docs

**7.** Tana Outliner - Supertags and typed nodes: https://outliner.tana.inc/learn/features/supertags

**8.** Xmind User Guide - outline, Zen, and Pitch modes: https://xmind.com/user-guide/intro-new

**9.** Miro Help Center - diagrams and mind maps: https://help.miro.com/hc/en-us/articles/28782102127890-Miro-AI-with-Diagrams-and-mindmaps

**10.** Heptabase Help Center - whiteboards, cards, mindmaps, and knowledge context: https://support.heptabase.com/

**11.** Community discussions examined included quick-capture friction, local ownership, mobile task complexity, and handwriting versus typing trade-offs on Reddit note-taking communities.

# 29. Change Log

| Version | Date | Change |
|---|---|---|
| 0.1 | 24 July 2026 | Initial canonical product vision. |
| 0.2 | 24 July 2026 | Added research-backed journal and mind-map design, visual design bible, onboarding/adoption rules, same-data multi-view architecture, accessibility, non-functional requirements, expanded acceptance criteria, and implementation sequencing. |